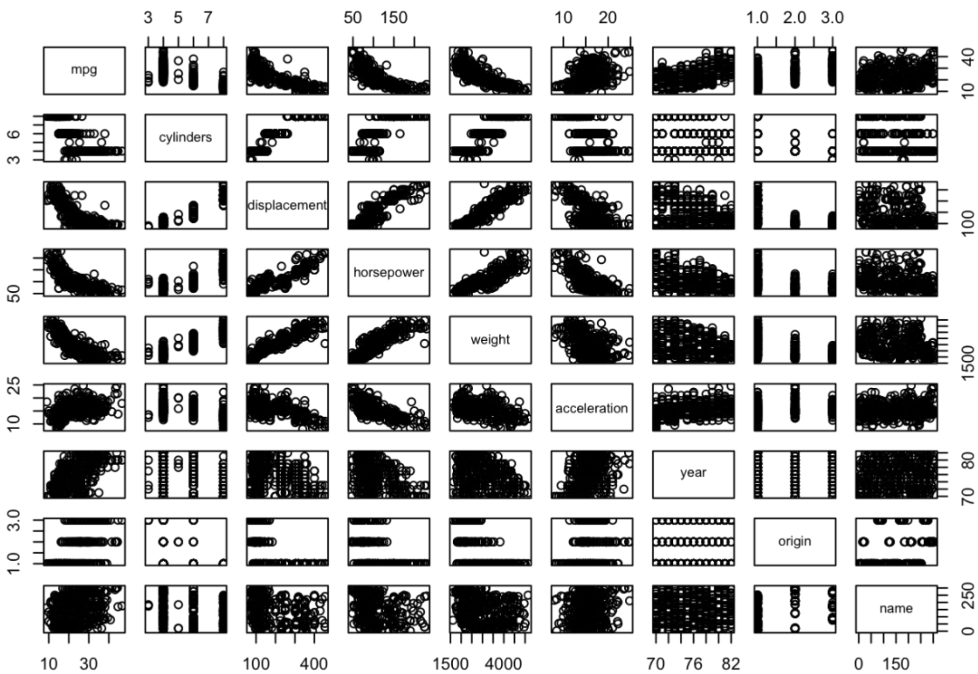
Chapter 3

9a.

```
pairs(Auto)
```



9b.

```
cor(subset(Auto, select=-name))
##                  mpg cylinders displacement horsepower  weight
## mpg           1.0000   -0.7776      -0.8051    -0.7784 -0.8322
## cylinders    -0.7776    1.0000       0.9508     0.8430  0.8975
## displacement -0.8051    0.9508       1.0000     0.8973  0.9330
## horsepower   -0.7784    0.8430       0.8973     1.0000  0.8645
## weight       -0.8322    0.8975       0.9330     0.8645  1.0000
## acceleration  0.4233   -0.5047      -0.5438    -0.6892 -0.4168
## year          0.5805   -0.3456      -0.3699    -0.4164 -0.3091
## origin        0.5652   -0.5689      -0.6145    -0.4552 -0.5850
##              acceleration    year  origin
## mpg                0.4233  0.5805  0.5652
## cylinders         -0.5047 -0.3456 -0.5689
## displacement      -0.5438 -0.3699 -0.6145
## horsepower        -0.6892 -0.4164 -0.4552
## weight            -0.4168 -0.3091 -0.5850
## acceleration       1.0000  0.2903  0.2127
## year               0.2903  1.0000  0.1815
## origin             0.2127  0.1815  1.0000
```

9c.

```
lm.fit1 = lm(mpg~.-name, data=Auto)
summary(lm.fit1)
##
## Call:
## lm(formula = mpg ~ . - name, data = Auto)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -9.590 -2.157 -0.117  1.869 13.060
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1.72e+01   4.64e+00   -3.71  0.00024 ***
## cylinders    -4.93e-01   3.23e-01   -1.53  0.12780
## displacement  1.99e-02   7.51e-03    2.65  0.00844 **
## horsepower   -1.70e-02   1.38e-02   -1.23  0.21963
## weight       -6.47e-03   6.52e-04   -9.93  < 2e-16 ***
## acceleration  8.06e-02   9.88e-02    0.82  0.41548
## year          7.51e-01   5.10e-02   14.73  < 2e-16 ***
## origin        1.43e+00   2.78e-01    5.13  4.7e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.33 on 384 degrees of freedom
## Multiple R-squared:  0.821,  Adjusted R-squared:  0.818
## F-statistic:  252 on 7 and 384 DF,  p-value: <2e-16
```

i.

Yes, there is a relatioship between the predictors and the response by testing the null hypothesis of whether all the regression coefficients are zero. The F -statistic is far from 1 (with a small p-value), indicating evidence against the null hypothesis.

ii.

Looking at the p-values associated with each predictor’s t-statistic, we see that displacement, weight, year, and origin have a statistically significant relationship, while cylinders, horsepower, and acceleration do not.

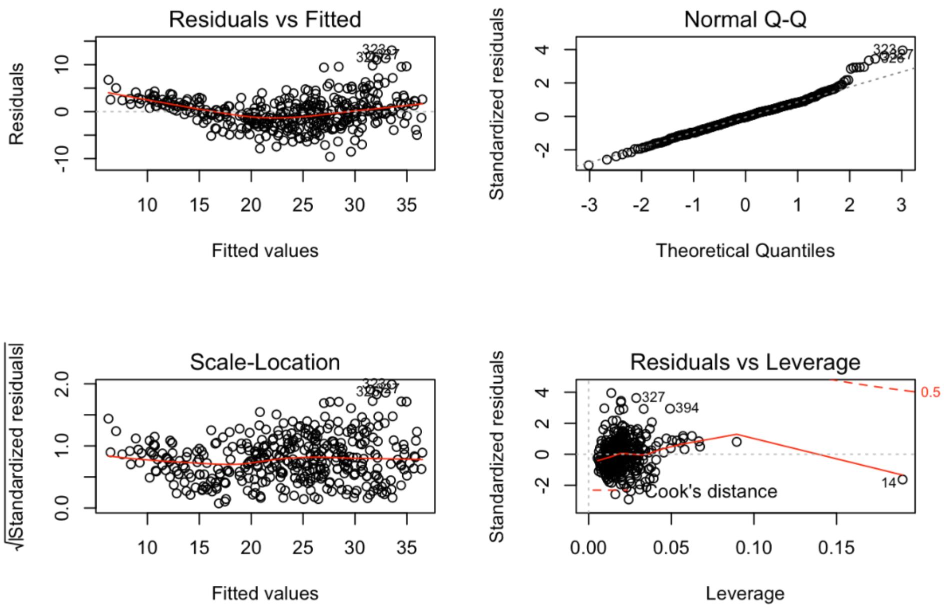
iii.

The regression coefficient for year, `0.7508`, suggests that for every one year, mpg increases by the coefficient. In other words, cars become more fuel efficient every year by almost 1 mpg / year.

9d.

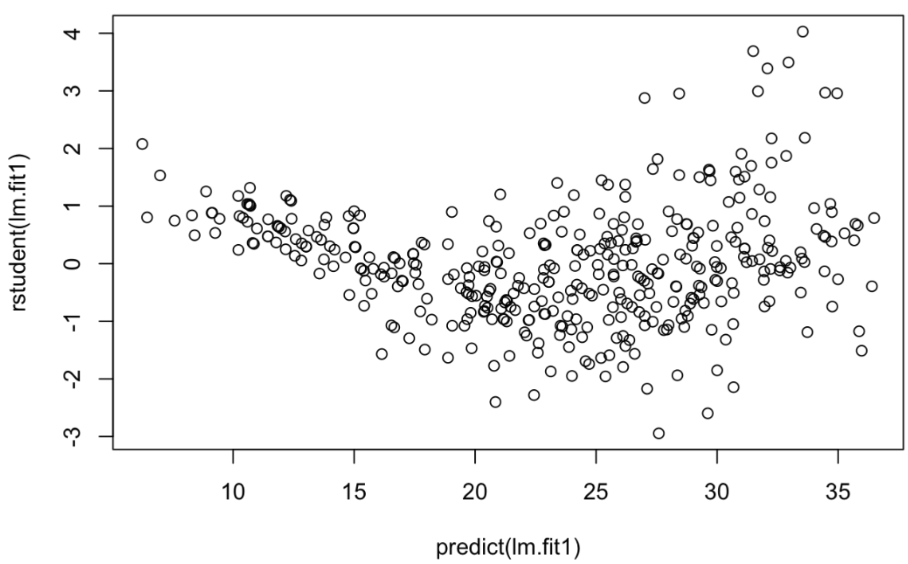
```
par(mfrow=c(2,2))
```

```
plot(lm.fit1)
```



The fit does not appear to be accurate because there is a discernible curve pattern to the residuals plots. From the leverage plot, point 14 appears to have high leverage, although not a high magnitude residual.

```
plot(predict(lm.fit1), rstudent(lm.fit1))
```



There are possible outliers as seen in the plot of studentized residuals because there are data with a value greater than 3.

9e.

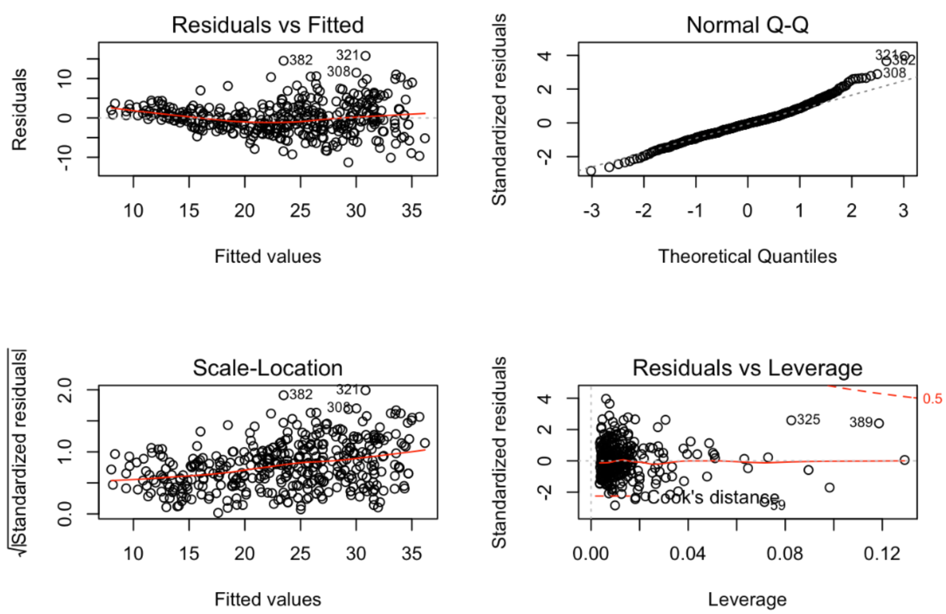
```
lm.fit2 = lm(mpg~cylinders*displacement+displacement*weight)
summary(lm.fit2)
## Call:
## lm(formula = mpg ~ cylinders * displacement + displacement *
##     weight)
## Residuals:
##     Min      1Q  Median      3Q     Max
## -13.293  -2.518  -0.348   1.840  17.772
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)             5.26e+01   2.24e+00   23.52  < 2e-16 ***
## cylinders               7.61e-01   7.67e-01    0.99     0.32
## displacement           -7.35e-02   1.67e-02   -4.40  1.4e-05 ***
## weight                 -9.89e-03   1.33e-03   -7.44  6.7e-13 ***
## cylinders:displacement -2.99e-03   3.43e-03   -0.87     0.38
## displacement:weight     2.13e-05   5.00e-06    4.25  2.6e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.1 on 386 degrees of freedom
## Multiple R-squared:  0.727,  Adjusted R-squared:  0.724
## F-statistic:  206 on 5 and 386 DF,  p-value: <2e-16
```

From the correlation matrix, I obtained the two highest correlated pairs and used them in picking my interaction effects. From the p-values, we can see that the interaction between displacement and weight is statistically signifcant, while the interactiion between cylinders and displacement is not.

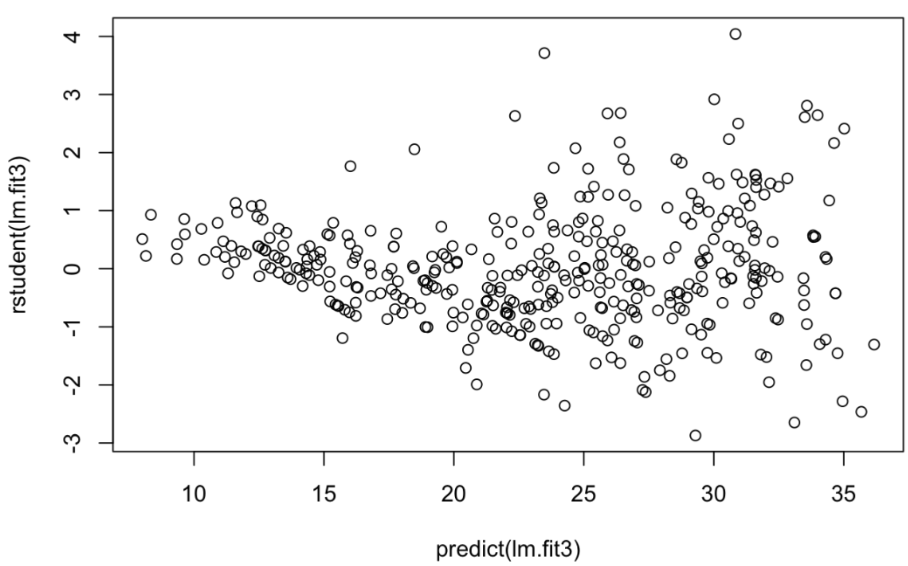
9f.

```
lm.fit3 = lm(mpg~log(weight)+sqrt(horsepower)+acceleration+I(acceleration^2))
summary(lm.fit3)
## Call:
## lm(formula = mpg ~ log(weight) + sqrt(horsepower) + acceleration +
##     I(acceleration^2))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -11.293  -2.508  -0.224   2.024  15.765
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)       178.3030    10.8045   16.50  < 2e-16 ***
## log(weight)       -14.7426     1.7399   -8.47  5.1e-16 ***
## sqrt(horsepower)   -1.8519     0.3600   -5.14  4.3e-07 ***
## acceleration       -2.1989     0.6390   -3.44  0.00064 ***
## I(acceleration^2)   0.0614     0.0186    3.31  0.00104 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 3.99 on 387 degrees of freedom
## Multiple R-squared:  0.741,  Adjusted R-squared:  0.739
## F-statistic:  277 on 4 and 387 DF,  p-value: <2e-16
par(mfrow=c(2,2))
plot(lm.fit3)
```




```
plot(predict(lm.fit3), rstudent(lm.fit3))
```



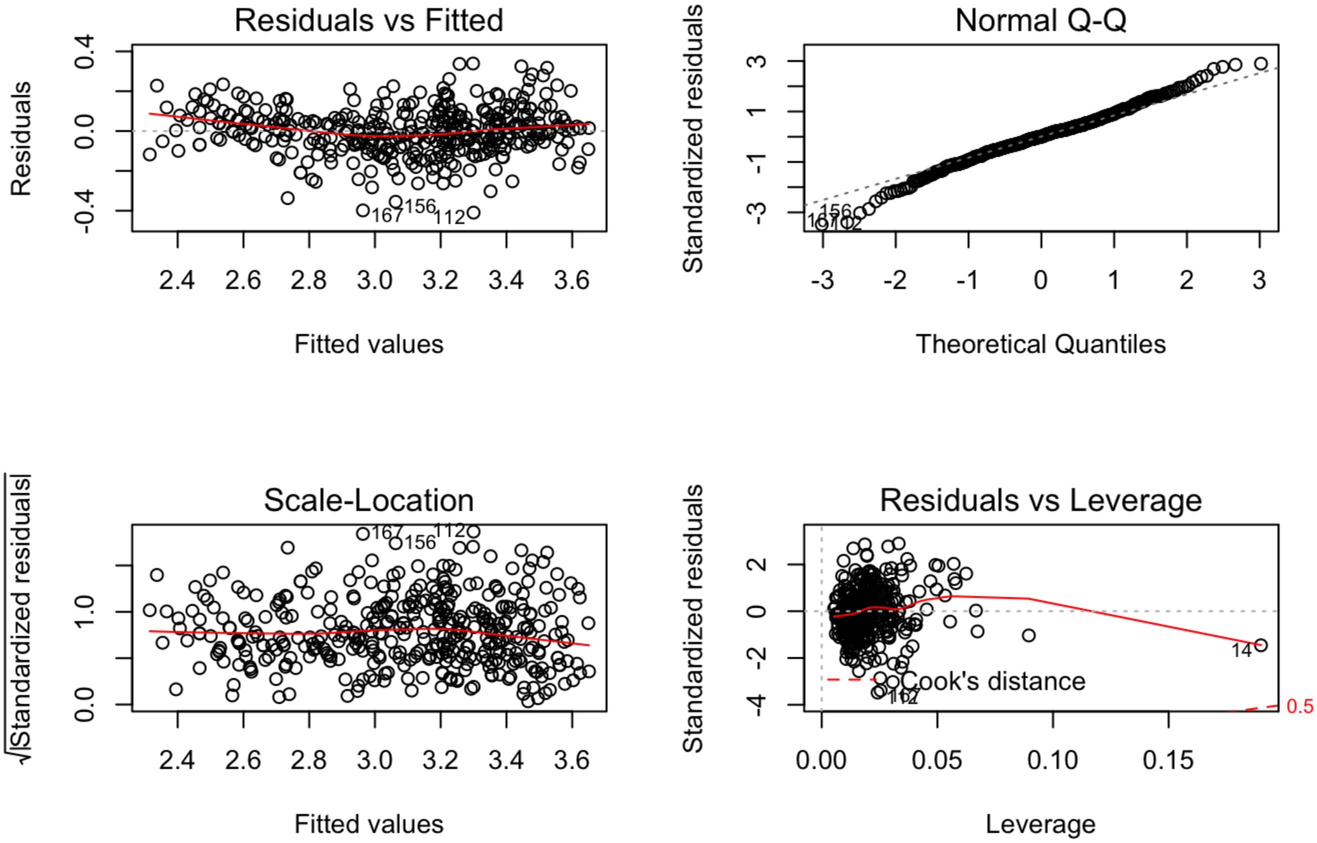
Apparently, from the p-values, the log(weight), sqrt(horsepower), and acceleration^2 all have statistical significance of some sort. The residuals plot has less of a discernible pattern than the plot of all linear regression terms. The studentized residuals displays potential outliers (>3). The leverage plot indicates more than three points with high leverage.

However, 2 problems are observed from the above plots: 1) the residuals vs fitted plot indicates heteroskedasticity (unconstant variance over mean) in the model. 2) The Q-Q plot indicates somewhat unnormality of the residuals.

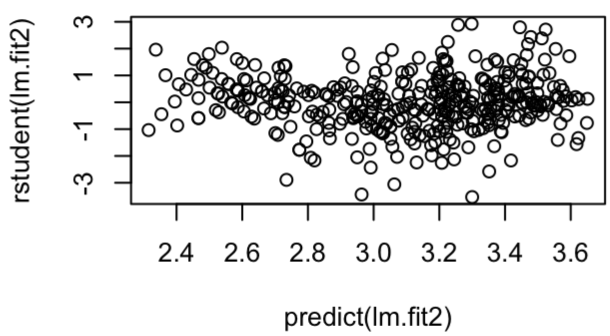
So, a better transformation need to be applied to our model. From the correlation matrix in 9a., displacement, horsepower and weight show a similar nonlinear pattern against our response mpg. This nonlinear pattern is very close to a log form. So in the next attempt, we use `log(mpg)` as our response variable.

The outputs show that log transform of mpg yield better model fitting (better R^2, normality of residuals).

```
lm.fit2<-
lm(log(mpg)~cylinders+displacement+horsepower+weight+acceleration+year+origin
,data=Auto)
summary(lm.fit2)
##
## Call:
## lm(formula = log(mpg) ~ cylinders + displacement + horsepower +
##     weight + acceleration + year + origin, data = Auto)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.4096 -0.0653  0.0008  0.0679  0.3392
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.75e+00   1.66e-01   10.53  < 2e-16 ***
## cylinders    -2.79e-02   1.16e-02   -2.42    0.016 *
## displacement  6.36e-04   2.69e-04    2.37    0.019 *
## horsepower   -1.47e-03   4.94e-04   -2.99    0.003 **
## weight       -2.55e-04   2.33e-05  -10.93  < 2e-16 ***
## acceleration -1.35e-03   3.54e-03   -0.38    0.703
## year          2.96e-02   1.82e-03   16.21  < 2e-16 ***
## origin        4.07e-02   9.96e-03    4.09  5.3e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.119 on 384 degrees of freedom
## Multiple R-squared:  0.88,   Adjusted R-squared:  0.877
## F-statistic:  400 on 7 and 384 DF,  p-value: <2e-16
par(mfrow=c(2,2))
plot(lm.fit2)
```




```
plot(predict(lm.fit2),rstudent(lm.fit2))
```



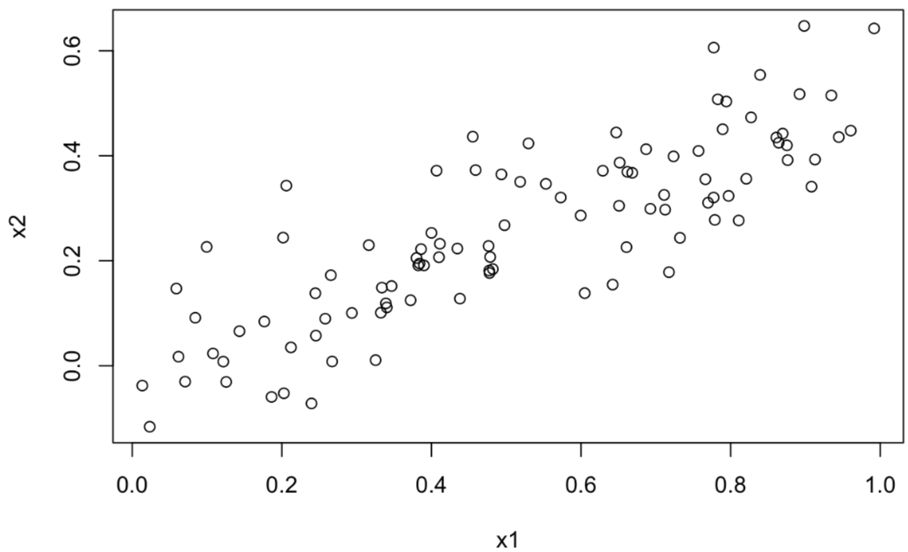
14a.

```
set.seed(1)
x1 = runif(100)
x2 = 0.5 * x1 + rnorm(100)/10
y = 2 + 2*x1 + 0.3*x2 + rnorm(100)
```

$$Y=2+2X_1+0.3X_2+\epsilon$$
$$\beta_0=2,\beta_1=2,\beta_3=0.3$$

14b.

```
cor(x1, x2)
## [1] 0.8351
plot(x1, x2)
```



14c.

```
lm.fit = lm(y~x1+x2)
summary(lm.fit)
##
## Call:
## lm(formula = y ~ x1 + x2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.8311 -0.7273 -0.0537  0.6338  2.3359
##
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.130      0.232    9.19  7.6e-15 ***
## x1             1.440      0.721    2.00    0.049 *
## x2             1.010      1.134    0.89    0.375
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.06 on 97 degrees of freedom
## Multiple R-squared:  0.209,  Adjusted R-squared:  0.193
## F-statistic: 12.8 on 2 and 97 DF,  p-value: 1.16e-05
```

$$\beta_0=2.0533,\beta_1=1.6336,\beta_3=0.5588$$β0=2.0533,β1=1.6336,β3=0.5588

The regression coefficients are close to the true coefficients, although with high standard error. We can reject the null hypothesis for β_1β1 because its p-value is below 5%. We cannot reject the null hypothesis for β_2β2 because its p-value is much above the 5% typical cutoff, over 60%.

14d.

```
lm.fit = lm(y~x1)
summary(lm.fit)
##
## Call:
## lm(formula = y ~ x1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.8950 -0.6687 -0.0779  0.5922  2.4556
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.112      0.231    9.15  8.3e-15 ***
## x1             1.976      0.396    4.99  2.7e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.06 on 98 degrees of freedom
## Multiple R-squared:  0.202,  Adjusted R-squared:  0.194
## F-statistic: 24.9 on 1 and 98 DF,  p-value: 2.66e-06
```

Yes, we can reject the null hypothesis for the regression coefficient given the p-value for its t-statistic is near zero.

14e.

```
lm.fit = lm(y~x2)
summary(lm.fit)
##
## Call:
## lm(formula = y ~ x2)
##
```

```
## Residuals:
##    Min     1Q Median     3Q    Max
## -2.627 -0.752 -0.036  0.724  2.449
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.390      0.195   12.26  < 2e-16 ***
## x2             2.900      0.633    4.58  1.4e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.07 on 98 degrees of freedom
## Multiple R-squared:  0.176,  Adjusted R-squared:  0.168
## F-statistic:   21 on 1 and 98 DF,  p-value: 1.37e-05
```

Yes, we can reject the null hypothesis for the regression coefficient given the p-value for its t-statistic is near zero.

14f.

No, because x1 and x2 have collinearity, it is hard to distinguish their effects when regressed upon together. When they are regressed upon separately, the linear relationship between y and each predictor is indicated more clearly.

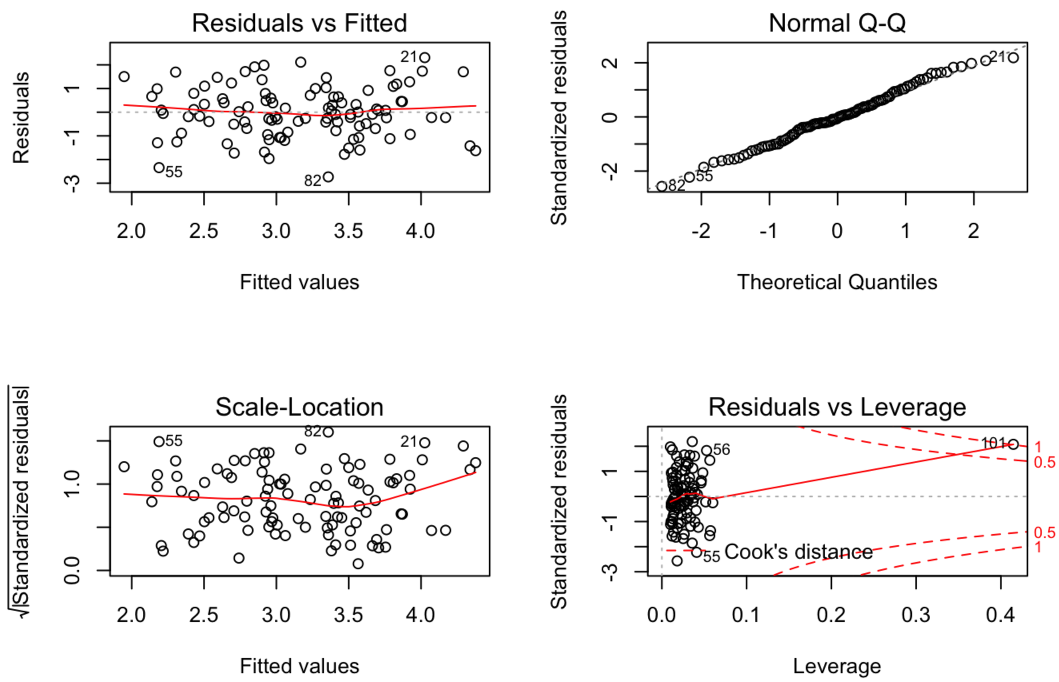
14g.

```
x1 = c(x1, 0.1)
x2 = c(x2, 0.8)
y = c(y, 6)
lm.fit1 = lm(y~x1+x2)
summary(lm.fit1)
##
## Call:
## lm(formula = y ~ x1 + x2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7335 -0.6932 -0.0526  0.6638  2.3062
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.227      0.231    9.62  7.9e-16 ***
## x1             0.539      0.592    0.91   0.3646
## x2             2.515      0.898    2.80   0.0061 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.07 on 98 degrees of freedom
## Multiple R-squared:  0.219,  Adjusted R-squared:  0.203
## F-statistic: 13.7 on 2 and 98 DF,  p-value: 5.56e-06
lm.fit2 = lm(y~x1)
summary(lm.fit2)
##
```

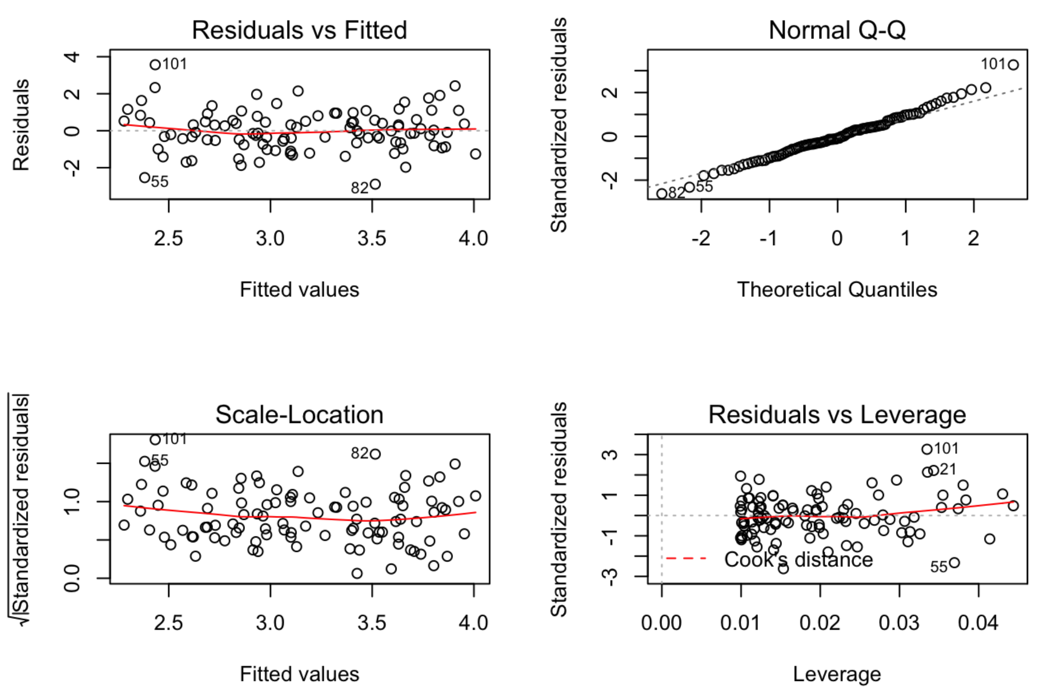
```
## Call:
## lm(formula = y ~ x1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.890 -0.656 -0.091  0.568  3.567
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.257      0.239    9.44  1.8e-15 ***
## x1             1.766      0.412    4.28  4.3e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.11 on 99 degrees of freedom
## Multiple R-squared:  0.156,  Adjusted R-squared:  0.148
## F-statistic: 18.3 on 1 and 99 DF,  p-value: 4.29e-05
lm.fit3 = lm(y~x2)
summary(lm.fit3)
##
## Call:
## lm(formula = y ~ x2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.647 -0.710 -0.069  0.727  2.381
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.345      0.191   12.26  < 2e-16 ***
## x2             3.119      0.604    5.16  1.3e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.07 on 99 degrees of freedom
## Multiple R-squared:  0.212,  Adjusted R-squared:  0.204
## F-statistic: 26.7 on 1 and 99 DF,  p-value: 1.25e-06
```

In the first model, it shifts x1 to statistically insignificance and shifts x2 to statistiscal significance from the change in p-values between the two linear regressions.

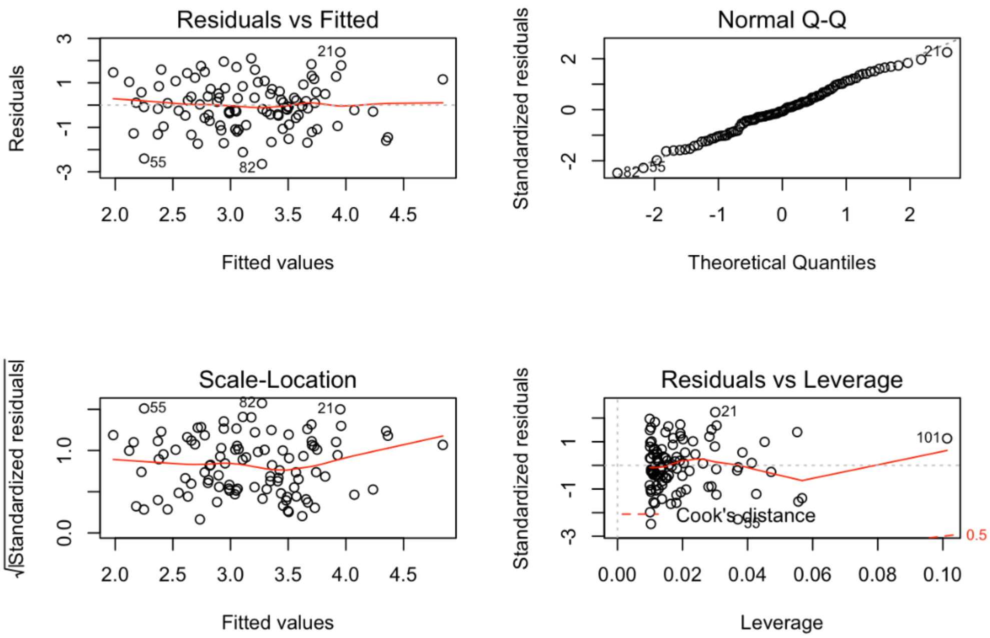
```
par(mfrow=c(2,2))
plot(lm.fit1)
```




```
par(mfrow=c(2,2))
plot(lm.fit2)
```

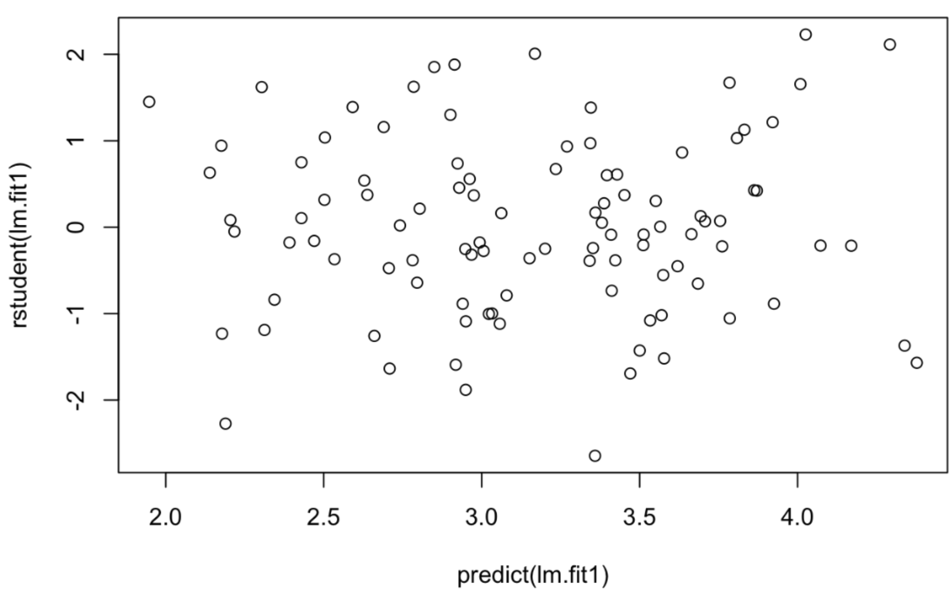


```
par(mfrow=c(2,2))
plot(lm.fit3)
```

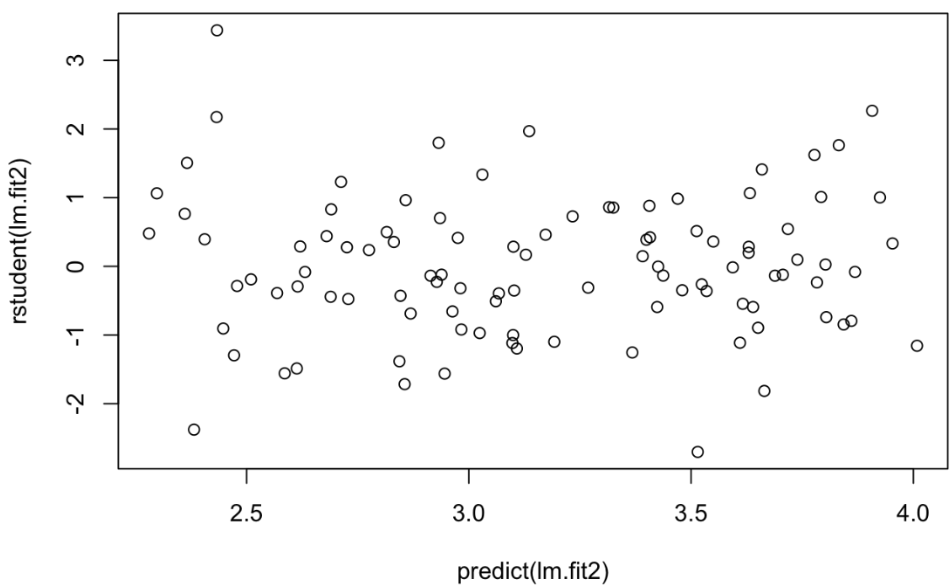


In the first and third models, the point becomes a high leverage point.

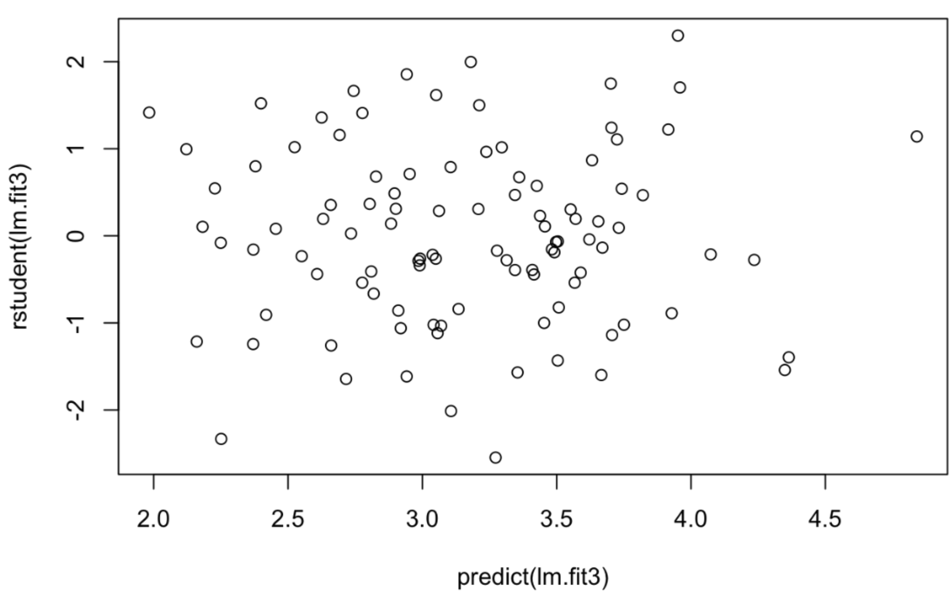
```
plot(predict(lm.fit1), rstudent(lm.fit1))
```



```
plot(predict(lm.fit2), rstudent(lm.fit2))
```




```
plot(predict(lm.fit3), rstudent(lm.fit3))
```



Looking at the studentized residuals, we don't observe points too far from the |3| value cutoff, except for the second linear regression: y ~ x1.

15a.

```
library(MASS)
## Warning: package 'MASS' was built under R version 3.0.2
summary(Boston)
##       crim              zn             indus            chas
##  Min.   : 0.01   Min.   :  0.0   Min.   : 0.46   Min.   :0.0000
##  1st Qu.: 0.08   1st Qu.:  0.0   1st Qu.: 5.19   1st Qu.:0.0000
##  Median : 0.26   Median :  0.0   Median : 9.69   Median :0.0000
##  Mean   : 3.61   Mean   : 11.4   Mean   :11.14   Mean   :0.0692
##  3rd Qu.: 3.68   3rd Qu.: 12.5   3rd Qu.:18.10   3rd Qu.:0.0000
##  Max.   :88.98   Max.   :100.0   Max.   :27.74   Max.   :1.0000
##       nox              rm             age             dis
##  Min.   :0.385   Min.   :3.56   Min.   :  2.9   Min.   : 1.13
##  1st Qu.:0.449   1st Qu.:5.89   1st Qu.: 45.0   1st Qu.: 2.10
##  Median :0.538   Median :6.21   Median : 77.5   Median : 3.21
##  Mean   :0.555   Mean   :6.29   Mean   : 68.6   Mean   : 3.79
##  3rd Qu.:0.624   3rd Qu.:6.62   3rd Qu.: 94.1   3rd Qu.: 5.19
##  Max.   :0.871   Max.   :8.78   Max.   :100.0   Max.   :12.13
##       rad              tax         ptratio         black
##  Min.   : 1.00   Min.   :187   Min.   :12.6   Min.   :  0.3
##  1st Qu.: 4.00   1st Qu.:279   1st Qu.:17.4   1st Qu.:375.4
##  Median : 5.00   Median :330   Median :19.1   Median :391.4
##  Mean   : 9.55   Mean   :408   Mean   :18.5   Mean   :356.7
##  3rd Qu.:24.00   3rd Qu.:666   3rd Qu.:20.2   3rd Qu.:396.2
##  Max.   :24.00   Max.   :711   Max.   :22.0   Max.   :396.9
##      lstat            medv
##  Min.   : 1.73   Min.   : 5.0
##  1st Qu.: 6.95   1st Qu.:17.0
##  Median :11.36   Median :21.2
##  Mean   :12.65   Mean   :22.5
##  3rd Qu.:16.95   3rd Qu.:25.0
##  Max.   :37.97   Max.   :50.0
Boston$chas <- factor(Boston$chas, labels = c("N","Y"))
summary(Boston)
##       crim              zn             indus       chas         nox
##  Min.   : 0.01   Min.   :  0.0   Min.   : 0.46   N:471   Min.   :0.385
##  1st Qu.: 0.08   1st Qu.:  0.0   1st Qu.: 5.19   Y: 35   1st Qu.:0.449
##  Median : 0.26   Median :  0.0   Median : 9.69           Median :0.538
##  Mean   : 3.61   Mean   : 11.4   Mean   :11.14           Mean   :0.555
##  3rd Qu.: 3.68   3rd Qu.: 12.5   3rd Qu.:18.10           3rd Qu.:0.624
##  Max.   :88.98   Max.   :100.0   Max.   :27.74           Max.   :0.871
##        rm            age              dis              rad
##  Min.   :3.56   Min.   :  2.9   Min.   : 1.13   Min.   : 1.00
##  1st Qu.:5.89   1st Qu.: 45.0   1st Qu.: 2.10   1st Qu.: 4.00
##  Median :6.21   Median : 77.5   Median : 3.21   Median : 5.00
##  Mean   :6.29   Mean   : 68.6   Mean   : 3.79   Mean   : 9.55
##  3rd Qu.:6.62   3rd Qu.: 94.1   3rd Qu.: 5.19   3rd Qu.:24.00
##  Max.   :8.78   Max.   :100.0   Max.   :12.13   Max.   :24.00
##       tax         ptratio          black           lstat
##  Min.   :187   Min.   :12.6   Min.   :  0.3   Min.   : 1.73
##  1st Qu.:279   1st Qu.:17.4   1st Qu.:375.4   1st Qu.: 6.95
##  Median :330   Median :19.1   Median :391.4   Median :11.36
##  Mean   :408   Mean   :18.5   Mean   :356.7   Mean   :12.65
##  3rd Qu.:666   3rd Qu.:20.2   3rd Qu.:396.2   3rd Qu.:16.95
##  Max.   :711   Max.   :22.0   Max.   :396.9   Max.   :37.97
```

```
##       medv
##  Min.   : 5.0
##  1st Qu.:17.0
##  Median :21.2
##  Mean   :22.5
##  3rd Qu.:25.0
##  Max.   :50.0
attach(Boston)
lm.zn = lm(crim~zn)
summary(lm.zn) # yes
##
## Call:
## lm(formula = crim ~ zn)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -4.43  -4.22  -2.62   1.25  84.52
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.4537     0.4172   10.67  < 2e-16 ***
## zn           -0.0739     0.0161   -4.59  5.5e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.44 on 504 degrees of freedom
## Multiple R-squared:  0.0402, Adjusted R-squared:  0.0383
## F-statistic: 21.1 on 1 and 504 DF,  p-value: 5.51e-06
lm.indus = lm(crim~indus)
summary(lm.indus) # yes
##
## Call:
## lm(formula = crim ~ indus)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -11.97  -2.70  -0.74   0.71  81.81
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.064      0.667   -3.09   0.0021 **
## indus          0.510      0.051    9.99   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.87 on 504 degrees of freedom
## Multiple R-squared:  0.165,  Adjusted R-squared:  0.164
## F-statistic: 99.8 on 1 and 504 DF,  p-value: <2e-16
lm.chas = lm(crim~chas)
summary(lm.chas) # no
##
## Call:
## lm(formula = crim ~ chas)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -3.74  -3.66  -3.44   0.02  85.23
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.744      0.396    9.45   <2e-16 ***
## chasY         -1.893      1.506   -1.26     0.21
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.6 on 504 degrees of freedom
## Multiple R-squared:  0.00312,    Adjusted R-squared:  0.00115
## F-statistic: 1.58 on 1 and 504 DF,  p-value: 0.209
lm.nox = lm(crim~nox)
summary(lm.nox) # yes
##
## Call:
## lm(formula = crim ~ nox)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -12.37  -2.74  -0.97   0.56  81.73
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -13.7        1.7   -8.07  5.1e-15 ***
## nox             31.2        3.0   10.42  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.81 on 504 degrees of freedom
## Multiple R-squared:  0.177,  Adjusted R-squared:  0.176
## F-statistic:  109 on 1 and 504 DF,  p-value: <2e-16
lm.rm = lm(crim~rm)
summary(lm.rm) # yes
##
## Call:
## lm(formula = crim ~ rm)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -6.60  -3.95  -2.65   0.99  87.20
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   20.482      3.364    6.09  2.3e-09 ***
## rm            -2.684      0.532   -5.04  6.3e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.4 on 504 degrees of freedom
## Multiple R-squared:  0.0481, Adjusted R-squared:  0.0462
## F-statistic: 25.5 on 1 and 504 DF,  p-value: 6.35e-07
lm.age = lm(crim~age)
summary(lm.age) # yes
##
## Call:
## lm(formula = crim ~ age)
##
```

```
## Residuals:
##    Min     1Q Median     3Q    Max
##  -6.79  -4.26  -1.23   1.53  82.85
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -3.7779     0.9440   -4.00  7.2e-05 ***
## age           0.1078     0.0127    8.46  2.9e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.06 on 504 degrees of freedom
## Multiple R-squared:  0.124,  Adjusted R-squared:  0.123
## F-statistic: 71.6 on 1 and 504 DF,  p-value: 2.85e-16
lm.dis = lm(crim~dis)
summary(lm.dis) # yes
##
## Call:
## lm(formula = crim ~ dis)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -6.71  -4.13  -1.53   1.52  81.67
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    9.499      0.730   13.01   <2e-16 ***
## dis           -1.551      0.168   -9.21   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.97 on 504 degrees of freedom
## Multiple R-squared:  0.144,  Adjusted R-squared:  0.142
## F-statistic: 84.9 on 1 and 504 DF,  p-value: <2e-16
lm.rad = lm(crim~rad)
summary(lm.rad) # yes
##
## Call:
## lm(formula = crim ~ rad)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -10.16  -1.38  -0.14   0.66  76.43
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -2.2872     0.4435   -5.16  3.6e-07 ***
## rad           0.6179     0.0343   18.00  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.72 on 504 degrees of freedom
## Multiple R-squared:  0.391,  Adjusted R-squared:  0.39
## F-statistic:  324 on 1 and 504 DF,  p-value: <2e-16
lm.tax = lm(crim~tax)
summary(lm.tax) # yes
##
```

```
## Call:
## lm(formula = crim ~ tax)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -12.51  -2.74  -0.19   1.07  77.70
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.52837    0.81581   -10.4   <2e-16 ***
## tax          0.02974    0.00185    16.1   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7 on 504 degrees of freedom
## Multiple R-squared:  0.34,   Adjusted R-squared:  0.338
## F-statistic:  259 on 1 and 504 DF,  p-value: <2e-16
lm.ptratio = lm(crim~ptratio)
summary(lm.ptratio) # yes
##
## Call:
## lm(formula = crim ~ ptratio)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -7.65  -3.99  -1.91   1.82  83.35
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -17.647      3.147   -5.61  3.4e-08 ***
## ptratio        1.152      0.169    6.80  2.9e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.24 on 504 degrees of freedom
## Multiple R-squared:  0.0841, Adjusted R-squared:  0.0823
## F-statistic: 46.3 on 1 and 504 DF,  p-value: 2.94e-11
lm.black = lm(crim~black)
summary(lm.black) # yes
##
## Call:
## lm(formula = crim ~ black)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -13.76  -2.30  -2.09  -1.30  86.82
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 16.55353    1.42590   11.61   <2e-16 ***
## black       -0.03628    0.00387   -9.37   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.95 on 504 degrees of freedom
## Multiple R-squared:  0.148,  Adjusted R-squared:  0.147
## F-statistic: 87.7 on 1 and 504 DF,  p-value: <2e-16
```

```
lm.lstat = lm(crim~lstat)
summary(lm.lstat) # yes
##
## Call:
## lm(formula = crim ~ lstat)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -13.93  -2.82  -0.66   1.08  82.86
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -3.3305     0.6938    -4.8  2.1e-06 ***
## lstat         0.5488     0.0478    11.5  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.66 on 504 degrees of freedom
## Multiple R-squared:  0.208,  Adjusted R-squared:  0.206
## F-statistic:  132 on 1 and 504 DF,  p-value: <2e-16
lm.medv = lm(crim~medv)
summary(lm.medv) # yes
##
## Call:
## lm(formula = crim ~ medv)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -9.07  -4.02  -2.34   1.30  80.96
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  11.7965     0.9342   12.63   <2e-16 ***
## medv         -0.3632     0.0384   -9.46   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.93 on 504 degrees of freedom
## Multiple R-squared:  0.151,  Adjusted R-squared:  0.149
## F-statistic: 89.5 on 1 and 504 DF,  p-value: <2e-16
```

All, except chas. Plot each linear regression using “plot(lm)” to see residuals.

15b.

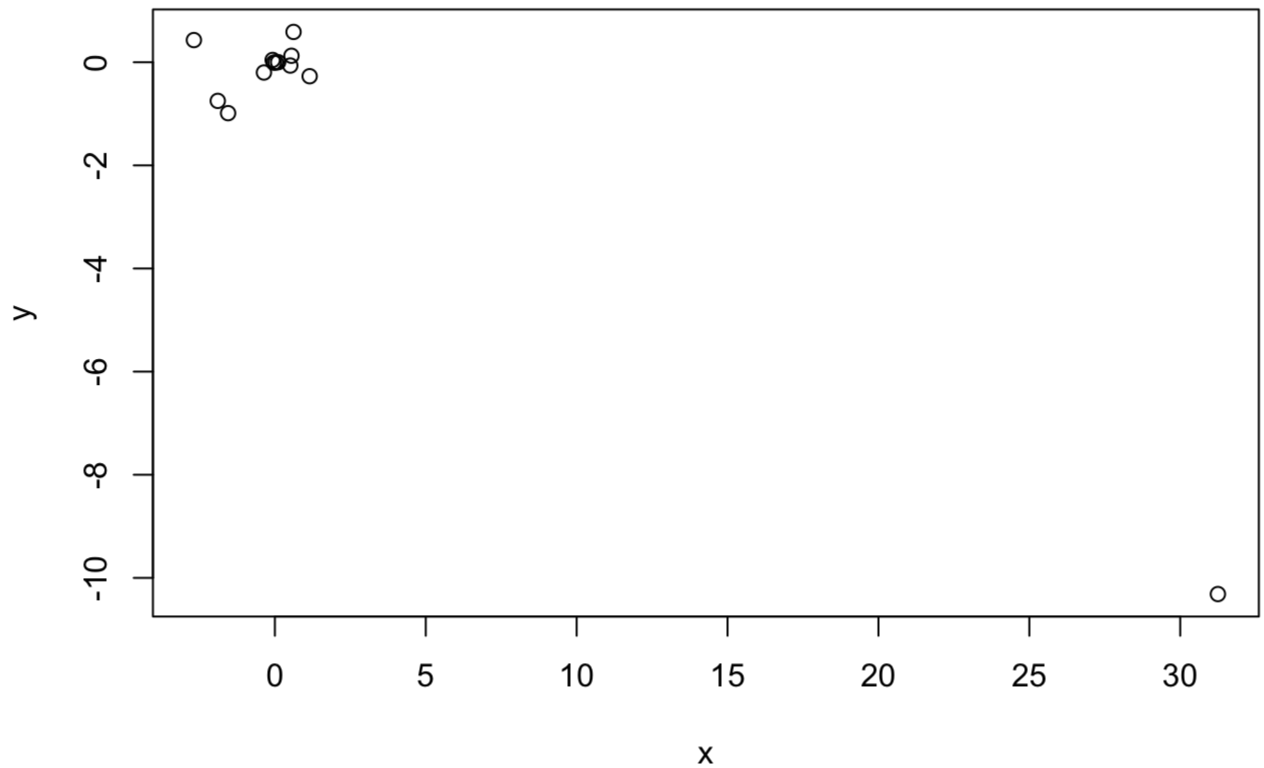
```
lm.all = lm(crim~., data=Boston)
summary(lm.all)
##
## Call:
## lm(formula = crim ~ ., data = Boston)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -9.92  -2.12  -0.35   1.02  75.05
##
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  17.03323    7.23490    2.35   0.0189 *
## zn            0.04486    0.01873    2.39   0.0170 *
## indus        -0.06385    0.08341   -0.77   0.4443
## chasY        -0.74913    1.18015   -0.63   0.5259
## nox         -10.31353    5.27554   -1.95   0.0512 .
## rm            0.43013    0.61283    0.70   0.4831
## age           0.00145    0.01793    0.08   0.9355
## dis          -0.98718    0.28182   -3.50   0.0005 ***
## rad           0.58821    0.08805    6.68  6.5e-11 ***
## tax          -0.00378    0.00516   -0.73   0.4638
## ptratio      -0.27108    0.18645   -1.45   0.1466
## black        -0.00754    0.00367   -2.05   0.0407 *
## lstat         0.12621    0.07572    1.67   0.0962 .
## medv         -0.19889    0.06052   -3.29   0.0011 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.44 on 492 degrees of freedom
## Multiple R-squared:  0.454,  Adjusted R-squared:  0.44
## F-statistic: 31.5 on 13 and 492 DF,  p-value: <2e-16
```

zn, dis, rad, black, medv

15c.

```
x = c(coefficients(lm.zn)[2],
      coefficients(lm.indus)[2],
      coefficients(lm.chas)[2],
      coefficients(lm.nox)[2],
      coefficients(lm.rm)[2],
      coefficients(lm.age)[2],
      coefficients(lm.dis)[2],
      coefficients(lm.rad)[2],
      coefficients(lm.tax)[2],
      coefficients(lm.ptratio)[2],
      coefficients(lm.black)[2],
      coefficients(lm.lstat)[2],
      coefficients(lm.medv)[2])
y = coefficients(lm.all)[2:14]
plot(x, y)
```



Coefficient for nox is approximately -10 in univariate model and 31 in multiple regression model.

15d.

```
lm.zn = lm(crim~poly(zn,3))
summary(lm.zn) # 1, 2
##
## Call:
## lm(formula = crim ~ poly(zn, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -4.82  -4.61  -1.29   0.47  84.13
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.614      0.372    9.71  < 2e-16 ***
## poly(zn, 3)1  -38.750      8.372   -4.63  4.7e-06 ***
## poly(zn, 3)2   23.940      8.372    2.86   0.0044 **
## poly(zn, 3)3  -10.072      8.372   -1.20   0.2295
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 8.37 on 502 degrees of freedom
## Multiple R-squared:  0.0582, Adjusted R-squared:  0.0526
## F-statistic: 10.3 on 3 and 502 DF,  p-value: 1.28e-06
lm.indus = lm(crim~poly(indus,3))
summary(lm.indus) # 1, 2, 3
##
## Call:
## lm(formula = crim ~ poly(indus, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -8.28  -2.51   0.05   0.76  79.71
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)          3.61       0.33   10.95  < 2e-16 ***
## poly(indus, 3)1    78.59       7.42   10.59  < 2e-16 ***
## poly(indus, 3)2   -24.39       7.42   -3.29   0.0011 **
## poly(indus, 3)3   -54.13       7.42   -7.29  1.2e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.42 on 502 degrees of freedom
## Multiple R-squared:  0.26,   Adjusted R-squared:  0.255
## F-statistic: 58.7 on 3 and 502 DF,  p-value: <2e-16
# lm.chas = lm(crim~poly(chas,3)) : qualitative predictor
lm.nox = lm(crim~poly(nox,3))
summary(lm.nox) # 1, 2, 3
##
## Call:
## lm(formula = crim ~ poly(nox, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -9.11  -2.07  -0.25   0.74  78.30
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.614      0.322   11.24  < 2e-16 ***
## poly(nox, 3)1   81.372      7.234   11.25  < 2e-16 ***
## poly(nox, 3)2  -28.829      7.234   -3.99  7.7e-05 ***
## poly(nox, 3)3  -60.362      7.234   -8.34  7.0e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.23 on 502 degrees of freedom
## Multiple R-squared:  0.297,  Adjusted R-squared:  0.293
## F-statistic: 70.7 on 3 and 502 DF,  p-value: <2e-16
lm.rm = lm(crim~poly(rm,3))
summary(lm.rm) # 1, 2
##
## Call:
## lm(formula = crim ~ poly(rm, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
```

```
## -18.49  -3.47  -2.22  -0.01  87.22
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)       3.61       0.37    9.76  < 2e-16 ***
## poly(rm, 3)1    -42.38       8.33   -5.09  5.1e-07 ***
## poly(rm, 3)2     26.58       8.33    3.19   0.0015 **
## poly(rm, 3)3     -5.51       8.33   -0.66   0.5086
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.33 on 502 degrees of freedom
## Multiple R-squared:  0.0678, Adjusted R-squared:  0.0622
## F-statistic: 12.2 on 3 and 502 DF,  p-value: 1.07e-07
lm.age = lm(crim~poly(age,3))
summary(lm.age) # 1, 2, 3
##
## Call:
## lm(formula = crim ~ poly(age, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -9.76  -2.67  -0.52   0.02  82.84
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)        3.614      0.349   10.37  < 2e-16 ***
## poly(age, 3)1     68.182      7.840    8.70  < 2e-16 ***
## poly(age, 3)2     37.484      7.840    4.78  2.3e-06 ***
## poly(age, 3)3     21.353      7.840    2.72   0.0067 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.84 on 502 degrees of freedom
## Multiple R-squared:  0.174,  Adjusted R-squared:  0.169
## F-statistic: 35.3 on 3 and 502 DF,  p-value: <2e-16
lm.dis = lm(crim~poly(dis,3))
summary(lm.dis) # 1, 2, 3
##
## Call:
## lm(formula = crim ~ poly(dis, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -10.76  -2.59   0.03   1.27  76.38
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)        3.614      0.326   11.09  < 2e-16 ***
## poly(dis, 3)1    -73.389      7.331  -10.01  < 2e-16 ***
## poly(dis, 3)2     56.373      7.331    7.69  7.9e-14 ***
## poly(dis, 3)3    -42.622      7.331   -5.81  1.1e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.33 on 502 degrees of freedom
## Multiple R-squared:  0.278,  Adjusted R-squared:  0.274
```

```
## F-statistic: 64.4 on 3 and 502 DF,  p-value: <2e-16
lm.rad = lm(crim~poly(rad,3))
summary(lm.rad) # 1, 2
##
## Call:
## lm(formula = crim ~ poly(rad, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -10.38  -0.41  -0.27   0.18  76.22
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)       3.614      0.297   12.16   <2e-16 ***
## poly(rad, 3)1  120.907      6.682   18.09   <2e-16 ***
## poly(rad, 3)2   17.492      6.682    2.62   0.0091 **
## poly(rad, 3)3    4.698      6.682    0.70   0.4823
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.68 on 502 degrees of freedom
## Multiple R-squared:   0.4,   Adjusted R-squared:  0.396
## F-statistic:  112 on 3 and 502 DF,  p-value: <2e-16
lm.tax = lm(crim~poly(tax,3))
summary(lm.tax) # 1, 2
##
## Call:
## lm(formula = crim ~ poly(tax, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -13.27  -1.39   0.05   0.54  76.95
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)       3.614      0.305   11.86  < 2e-16 ***
## poly(tax, 3)1  112.646      6.854   16.44  < 2e-16 ***
## poly(tax, 3)2   32.087      6.854    4.68  3.7e-06 ***
## poly(tax, 3)3   -7.997      6.854   -1.17     0.24
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.85 on 502 degrees of freedom
## Multiple R-squared:  0.369,  Adjusted R-squared:  0.365
## F-statistic: 97.8 on 3 and 502 DF,  p-value: <2e-16
lm.ptratio = lm(crim~poly(ptratio,3))
summary(lm.ptratio) # 1, 2, 3
##
## Call:
## lm(formula = crim ~ poly(ptratio, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -6.83  -4.15  -1.65   1.41  82.70
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)          3.614      0.361   10.01  < 2e-16 ***
## poly(ptratio, 3)1   56.045      8.122    6.90  1.6e-11 ***
## poly(ptratio, 3)2   24.775      8.122    3.05   0.0024 **
## poly(ptratio, 3)3  -22.280      8.122   -2.74   0.0063 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.12 on 502 degrees of freedom
## Multiple R-squared:  0.114,  Adjusted R-squared:  0.108
## F-statistic: 21.5 on 3 and 502 DF,  p-value: 4.17e-13
lm.black = lm(crim~poly(black,3))
summary(lm.black) # 1
##
## Call:
## lm(formula = crim ~ poly(black, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -13.10  -2.34  -2.13  -1.44  86.79
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)        3.614      0.354   10.22   <2e-16 ***
## poly(black, 3)1  -74.431      7.955   -9.36   <2e-16 ***
## poly(black, 3)2    5.926      7.955    0.75     0.46
## poly(black, 3)3   -4.835      7.955   -0.61     0.54
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.95 on 502 degrees of freedom
## Multiple R-squared:  0.15,   Adjusted R-squared:  0.145
## F-statistic: 29.5 on 3 and 502 DF,  p-value: <2e-16
lm.lstat = lm(crim~poly(lstat,3))
summary(lm.lstat) # 1, 2
##
## Call:
## lm(formula = crim ~ poly(lstat, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -15.23  -2.15  -0.49   0.07  83.35
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)        3.614      0.339   10.65   <2e-16 ***
## poly(lstat, 3)1   88.070      7.629   11.54   <2e-16 ***
## poly(lstat, 3)2   15.888      7.629    2.08    0.038 *
## poly(lstat, 3)3  -11.574      7.629   -1.52    0.130
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.63 on 502 degrees of freedom
## Multiple R-squared:  0.218,  Adjusted R-squared:  0.213
## F-statistic: 46.6 on 3 and 502 DF,  p-value: <2e-16
lm.medv = lm(crim~poly(medv,3))
summary(lm.medv) # 1, 2, 3
##
```

```
## Call:
## lm(formula = crim ~ poly(medv, 3))
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -24.43  -1.98  -0.44   0.44  73.65
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)        3.614      0.292   12.37   <2e-16 ***
## poly(medv, 3)1  -75.058      6.569  -11.43   <2e-16 ***
## poly(medv, 3)2   88.086      6.569   13.41   <2e-16 ***
## poly(medv, 3)3  -48.033      6.569   -7.31    1e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.57 on 502 degrees of freedom
## Multiple R-squared:  0.42,   Adjusted R-squared:  0.417
## F-statistic:  121 on 3 and 502 DF,  p-value: <2e-16
```

See inline comments above, the answer is yes for most, except for black and chas.

Chapter 4

6.

$$p(X) = \frac{\exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2)}{1 + \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2)}$$
$$X_1 = hoursstudied, X_2 = undergradGPA$$
$$\beta_0 = -6, \beta_1 = 0.05, \beta_2 = 1$$

a.

$$X = [40hours, 3.5GPA]$$
$$p(X) = \frac{\exp(-6 + 0.05X_1 + X_2)}{1 + \exp(-6 + 0.05X_1 + X_2)}$$
$$= \frac{\exp(-6 + 0.0540 + 3.5)}{1 + \exp(-6 + 0.0540 + 3.5)}$$
$$= \frac{\exp(-0.5)}{1 + \exp(-0.5)}$$
$$= 37.75\%$$

b.

$$X = [X_1 hours, 3.5GPA]$$
$$p(X) = \frac{\exp(-6 + 0.05X_1 + X_2)}{1 + \exp(-6 + 0.05X_1 + X_2)}$$
$$0.50 = \frac{\exp(-6 + 0.05X_1 + 3.5)}{1 + \exp(-6 + 0.05X_1 + 3.5)}$$
$$0.50(1 + \exp(-2.5 + 0.05X_1)) = \exp(-2.5 + 0.05X_1)$$
$$0.50 + 0.50\exp(-2.5 + 0.05X_1)) = \exp(-2.5 + 0.05X_1)$$
$$0.50 = 0.50\exp(-2.5 + 0.05X_1)$$
$$\log(1) = -2.5 + 0.05X_1$$
$$X_1 = 2.5/0.05 = 50hours$$

15.

a

```
Power = function() {
    2^3
}
print(Power())
## [1] 8
```

b

```
Power2 = function(x, a) {
    x^a
}
Power2(3, 8)
## [1] 6561
```

c

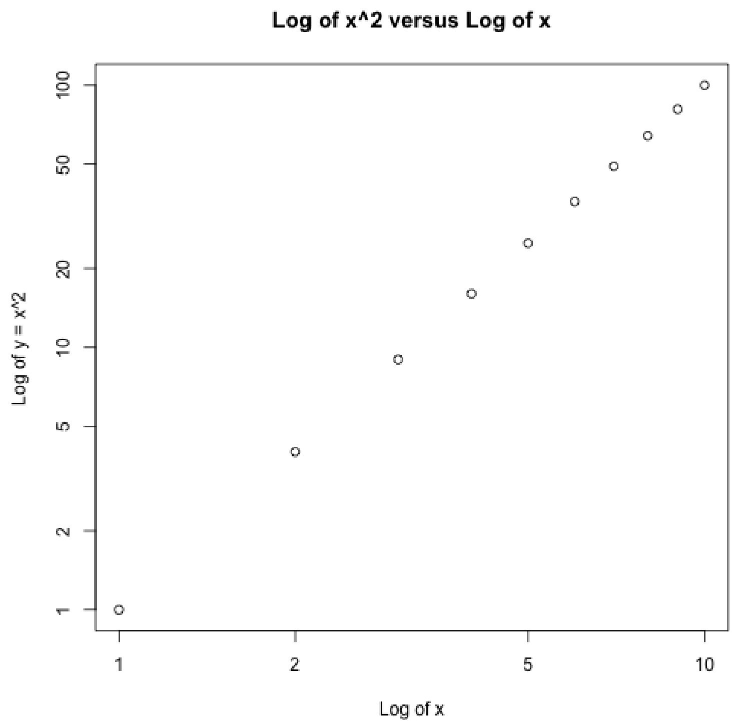
```
Power2(10, 3)
## [1] 1000
Power2(8, 17)
## [1] 2.252e+15
Power2(131, 3)
## [1] 2248091
```

d

```
Power3 = function(x, a) {
    result = x^a
    return(result)
}
```

e

```
x = 1:10
plot(x, Power3(x, 2), log = "xy", ylab = "Log of y = x^2", xlab = "Log of x",
    main = "Log of x^2 versus Log of x")
```



f

```
PlotPower = function(x, a) {
    plot(x, Power3(x, a))
}
PlotPower(1:10, 3)
```